Surya M S

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RESEARCH INTERESTS:

- VLSI and Analog IC Design
- Neuromorphic Computing and STDP Learning
- Low-Power and Energy-Efficient Circuits

EDUCATION:

Bachelor of Engineering – Electronics & Communication Engineering. Dec.22 – June*26*
Acharya Institute of Technology, India*(National Institutional Ranking Framework- Engineering Category: 23rd)*
CGPA: 7.17 (till 6th sem)
"Interested in VLSI and Analog IC Design with a focus on low-power, energy-efficient circuits. Motivated to explore neuromorphic computing and advanced semiconductor technologies for high-performance systems."

PROJECTS:

Optimized LIF Neuron with Memristor-Based Synapse and STDP Learning. *Dec.24 – 25*

- Designed and simulated a biologically inspired LIF neuron using CMOS technology and developed a memristor device model in Verilog-A for synaptic behavior in neuromorphic systems.
- Implemented memristor-based synapses between neurons and realized STDP learning using analog spike-timing circuits to achieve adaptive weight update through potentiation and depression.
- Performed pre-layout and post-layout verification of the neuron circuit, including DRC, LVS, parasitic extraction (Quantus), and post-layout simulations in 45 nm and 90 nm CMOS technologies.
- Analyzed power and energy efficiency across technology nodes, demonstrating significant reduction in power consumption at 45 nm compared to 90 nm for the integrated neuron–memristor system.

32-bit Vedic Multiplier using HSCG-SC Adder. Dec.24 – 25

- Architected high-speed 32-bit Vedic multiplier leveraging ancient mathematical principles for parallel processing.
- Integrated Hybrid Speed-Critical Carry-Generate/Sum-Carry (HSCG-SC) adder to optimize speed and minimize area overhead.
- Verified functionality, timing, and power via Xilinx simulations.

PUBLICATIONS & PRESENTATIONS:

Low-Power Neuromorphic Axon-Hillock Neuron with Memristor-Based Synapses for High-Efficiency Spiking Neural Networks. *Mar-Apr 2026*
Presented at an **international Scopus-indexed conference (IC-AISMART)**
Focused on low-power learning mechanisms and bio-inspired hardware models

TECHNICAL SKILLS:

- **Hardware & Design Tools**: Cadence Virtuoso, Xilinx ISE, LTspice, MATLAB
- **Programming / HDL**: Verilog, Python
- **Core Concepts**: VLSI Fundamentals, CMOS Design Basics, Neuromorphic Circuits

CERTIFICATIONS

- **NPTEL** – CMOS Digital VLSI Design *(8weeks)* ***Jan-Mar 2025***

ACHIEVEMENTS

- "State-level volleyball player with proven teamwork, leadership, and time-management skills gained through competitive sports participation."